

ARNAV GUPTA

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Experience

GetHelpDesk.ai

Jan 2026 – Present

Software Engineer | Python, Pipecat, FastAPI, LLMs

Remote

- Build and maintain a production LLM-powered voice-calling pipeline (Pipecat + FastAPI) serving real clients, owning reliability, incident response, and backend fixes end-to-end.
- Diagnosed failure modes in live conversational AI flows — traced root causes in LLM streaming and call-state handling — cutting production error rate by **90%** and failed calls by **20%**.
- Design and customise multi-step conversational agent workflows for client-specific use cases, translating ambiguous requirements into production-ready LLM-driven systems.

Varnan

May 2025 – July 2025

Web Development Intern | Next.js, TypeScript, Tailwind CSS

Remote

- Rebuilt the platform frontend in Next.js and integrated a CMS-backed content pipeline with RSS syndication, growing user reach by **30,000**.

Projects

Cognivia | FastAPI, LangChain, LLaMA (Ollama), Pinecone, MongoDB, Next.js, JWT

October 2025

- Built a full-stack AI e-learning platform with a RAG system over a self-hosted open-source LLM (Ollama/LLaMA), using Pinecone for vector search and LangChain to orchestrate retrieval chains.
- Engineered a heterogeneous ingestion pipeline — PDF text extraction and YouTube transcript aggregation — with semantic chunking and embedding, indexing **10,000+** documents for contextual retrieval.
- Auto-generates personalised study plans from user parameters using LLM reasoning over retrieved context; serves results through a FastAPI backend and a Next.js frontend with JWT authentication.

AI Social Media Automation | Python, FastAPI, Composio, LLM APIs

September 2025

- Built an LLM-powered automation system that transforms raw video scripts into platform-specific content for YouTube, Instagram, and Twitter via multi-step agent workflows orchestrated through Composio.
- Implemented an iterative LLM evaluation loop that rewrites posts against engagement heuristics, plus idempotent scheduled background jobs — reducing manual effort by **95%+**.

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, Java, C++

AI & LLM: LangChain, OpenAI API, Google Gemini API, Ollama (LLaMA), Pipecat (Voice AI), Composio, Prompt Engineering

RAG & Search: Retrieval-Augmented Generation, Semantic Chunking, Embeddings, Vector Search (Pinecone, Supabase)

ML & CV: Scikit-learn, NumPy, Pandas, NLTK, YOLOv8, OpenCV

Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST API design, JWT/OAuth

Frontend & DBs: React.js, Next.js, Tailwind CSS, PostgreSQL, MongoDB, MySQL, Redis

Tools: Git, GitHub, Docker, AWS S3, Vercel, Render, Postman

Education

Thapar Institute of Engineering and Technology

Aug. 2022 – May 2026

Bachelor of Engineering in Computer Science

Patiala, Punjab

- Relevant Coursework: Machine Learning, Natural Language Processing, Parallel & Distributed Programming, Data Structures & Algorithms

Leadership / Extracurricular

Microsoft Learn Student Chapter

April 2023 – June 2024

Coordinator

TIET

- Led a team of 10 to design and launch the official society website and Makeathon 6 event portal, driving a **150%** increase in user engagement.

Chemical Department

October 2023 – February 2024

UI/UX Design Head

TIET

- Designed UI/UX for the International Conference on Sustainable Development in Chemical Engineering, attracting **10,000+** global users.